Status	Finished
Started	Wednesday, 21 October 2026, 7:09 PM
Completed	Wednesday, 21 October 2026, 7:13 PM
Duration	4 mins 42 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it preserves the order of duplicate keys
- d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Heap sort
- c. Selection sort
- d. Quick sort

The correct answer is:
Merge sort

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Selection
c. Extraction
d. Insertion

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the largest element to its correct position
b. Sorting the entire list in one pass
c. Bubbling up the smallest element
d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **5**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sorted()
c. sort()
d. order()

The correct answer is: sorted()

Question **6**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It has no effect on the analysis process
- c. It makes the analysis slower
- d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **7**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It only works on small datasets
- c. It avoids the need for recursion
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It only works with integer arrays
- c. It sorts data out of the box efficiently
- d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It is based on the divide-and-conquer approach
c. It does not require recursion
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **10**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. The parts are ignored
c. Each part is sorted independently
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are discarded
c. They are split again into smaller sublists
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Dividing the input into smaller subproblems
b. Sorting the entire list sequentially
c. Combining sorted sublists
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It divides the list into two halves, sorts each half, and then merges them
d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Bubble Sort
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort